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Preparation of lignin nanoparticles via ultra-fast microwave-assisted fractionation of lignocellulose using ternary deep eutectic solvents

Zhiyan Yan[1] | Zhongren Wang[2] | Yuehui Chen[1] | Chenrui Liu[1] | Yun Liu[1] | Ruijie Li[2] | Mengying Si[2,3] | Yan Shi[2,3]

[1]Department of Environmental Science and Engineering, College of Environment and Resources, Xiangtan University, Xiangtan, China

[2]School of Metallurgy and Environment, Central South University, Changsha, China

[3]Chinese National Engineering Research Center for Control & Treatment of Heavy Metal Pollution, Changsha, China

Correspondence
Yun Liu, Department of Environmental Science and Engineering, College of Environment and Resources, Xiangtan University, Xiangtan 411105, China.
Email: liuyunscut@163.com

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Abstract

Lignin separation from natural lignocellulose for the preparation of lignin nanoparticles (LNPs) is often challenging owing to the recalcitrant and complex structure of lignocellulose. This paper reports a strategy for the rapid synthesis of LNPs via microwave-assisted lignocellulose fractionation using ternary deep eutectic solvents (DESs). A novel ternary DES with strong hydrogen bonding was prepared using choline chloride, oxalic acid, and lactic acid in a 1:0.5:1 ratio. Efficient fractionation of rice straw (0.5 × 2.0 cm) (RS) was realized by the ternary DES under microwave irradiation (680 W) within only 4 min, and 63.4% of lignin could be separated from the RS to prepare LNPs with a high lignin purity (86.8%), an average particle size of 48–95 nm, and a narrow size distribution. The mechanism of lignin conversion was also investigated, which revealed that dissolved lignin aggregated into LNPs via π–π stacking interactions.

KEYWORDS
lignin nanoparticles, lignocellulose fractionation, microwave, ternary deep eutectic solvents

1 | INTRODUCTION

Natural lignocellulose consists of cellulose, hemicellulose, and lignin. Among these, lignin is a heterogeneous polymer with a three-dimensional network structure composed of three phenylpropane units connected by intermolecular ether and carbon–carbon bonds. It is the second most abundant natural renewable biopolymer after cellulose and is a common by-product of the paper and pulp industry. The majority of lignin (95%) is burnt as fuel for heat and power production, and only a small amount is isolated (Duarah et al., 2020) which seriously impacts the environment. Owing to the excellent properties and extensive applications of nanomaterials, various lignin nanoparticles (LNPs) have been synthesized and used in different fields such as drug delivery, ultraviolet absorbers, and hybrid nanocomposites (Si et al., 2018). One of the most effective approaches for realizing high-value utilization of lignin is lignin nanocrystallization.

The separation of lignin is an essential prerequisite for the utilization of natural lignocellulose. The conversion of lignin into valuable products can significantly promote the development of biorefineries. Several studies have reported lignin separation via lignocellulose fractionation using ionic liquids, and this clean process presents several advantages including a high separation efficiency and good recyclability (Carrozza et al., 2019; Mohtar et al., 2015). However, lignin separation using ionic liquids is usually accompanied by the dissolution of cellulose and hemicellulose, which not only reduces their recovery rates but also increases the difficulty of lignin purification. Furthermore, the high production costs of ionic liquids hinder their industrialization in lignin valorization (Mäki-Arvela et al., 2010; Shen et al., 2015). To that end, deep eutectic solvents (DESs), prepared by mixing hydrogen-bond donors and receptors in a definite molar ratio to form a binary eutectic mixture with a melting point lower than that of the original components, have emerged as

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excellent green solvents with characteristics similar to traditional ionic liquids (Tian et al., 2020); however, these solvents have lower toxicity, weaker volatility, better biocompatibility, and improved biodegradability compared with traditional ionic liquids (Liu et al., 2019; Smith et al., 2014).

Notably, DESs can selectively dissolve lignin and hemicellulose primarily connected by benzyl ester or benzyl ether bonds in lignocellulosic matrices while maintaining the integrity of cellulose (Liu et al., 2017). The hydrogen-bond donors in DESs can replace hydrogen atoms to form hydrogen bonds with oxygen atoms in benzyl esters or benzyl ether bonds, thus weakening the intermolecular hydrogen bonds between lignin, hemicellulose, and cellulose. Furthermore, DESs are solutions of acids and bases that catalyze the breaking of ester and ether bonds. The removal of lignin from lignocellulose after fractionation with DESs has attracted considerable attention over the past few years; however, the requirements of high reaction temperatures (120–160°C), long treatment times (more than 1 h), and the mechanical treatment of lignocellulosic biomass before reactions have significantly hindered the industrial applications of DESs (Ai et al., 2020; Wang et al., 2020; Xu et al., 2020).

Several studies have reported that the efficiency of lignocellulose fractionation using DESs can be effectively improved by the addition of an organic acid as an electron donor or an inorganic acid to provide active protons and acid sites based on a binary system (Chen et al., 2018; Q. Xia et al., 2018; W. Xing et al., 2018).

This paper reports the microwave-assisted fractionation of rice straw (RS) using a ternary DES prepared from choline chloride (ChCl), lactic acid (LA), and oxalic acid (OA). The physicochemical properties of the ternary DESs are characterized. The effects of different DESs and the reaction time on lignin separation and the depolymerization mechanism of lignin in the DESs are investigated. Further, LNPs are prepared from the lignin separated using DESs, and their physicochemical properties and formation mechanisms are analyzed. Finally, the recycling efficiency of the DESs is investigated. Thus, our study provides novel insights into lignocellulose utilization using ternary DESs and guides the design of an effective strategy for LNP preparation.

2 | MATERIALS AND METHODS

2.1 | Biomass and chemicals

The RS, purchased from Jining, was washed with deionized water and then dried with heated air at 60°C for 36 h. The RS was then clipped into bulks (2 cm). ChCl, LA, and OA were purchased from Macklin Biochemical Co., Ltd.

2.2 | DES preparation and characterization

ChCl was mixed with LA, OA, or both (Supporting Information: Table S1) and stirred at 60°C until a transparent solution was obtained. The obtained DESs were stored in a vacuum desiccator until further use and were characterized via ^{1}H nuclear magnetic resonance (NMR) on a Bruker Avance 400 MHz spectrometer (Bruker GmbH) and a 1760X Fourier transform infrared spectrometer (FTIR, NICOLET IS10, Thermo). The viscosities of the DESs at 25°C and 60°C were determined using an NDJ-8S viscosimeter (Nirun). Nile red (NR), 4-nitroaniline (NH_2), and N,N-diethyl-4-nitroaniline (NEt_2) were used to calculate the Kamlet–Taft parameters of the DESs. The abovementioned 3 dyes were dissolved in methanol to a final concentration of 1.0×10^{-3}mol/L. The mixture containing the dye solution (100 μL) and DES (3 mL) was transferred into a quartz cuvette and placed into an ultraviolet–visible (UV–vis) spectrometer (U-4100; Hitachi) to record the visible spectra (400–700 nm). The Kamlet–Taft parameters (α, β, and π) were calculated according to Equations (1–5):

$$E_{NR} = 1/\lambda_{max,NR}, \quad (1)$$

$$v = 1/\lambda_{max}, \quad (2)$$

$$\pi = 0.314 \times (27.52 - v_{NEt_2}), \quad (3)$$

$$\beta = 0.358 \times (31.10 - v_{NH_2}) - 1.125 \times \pi^2, \quad (4)$$

$$\alpha = (19.9657 - 1.0241\,\pi - v_{NR})/1.6078. \quad (5)$$

Here, λ_{max} denotes the wavelength corresponding to the maximum absorption of the dye, α denotes the hydrogen-bond acidity, β denotes the hydrogen-bond basicity, and π denotes the dipolarity/polarizability parameter.

2.3 | RS fractionation using DESs and LNP preparation

Bulk RS (2.0 g) was mixed with the DES (60 mL) at a solid-to-liquid ratio of 1:30 (g/mL) in a microwave digestion tank, which was placed in a microwave oven at a power of 680 W for 3–5 min. The tank was then removed and allowed to cool naturally to room temperature. The mixture was filtered through a multilayer gauze to separate the solution and RS residue. To promote lignin aggregation, a 25-fold volume of water was then added to the solution, which was subsequently centrifuged at a centrifugal force of 24,000*g*. The LNPs were freeze-dried and stored in a desiccator. The RS residues were thoroughly washed with deionized water and air-dried for further measurements.

2.4 | Determination of the change in RS after DES treatment

The chemical compositions of the untreated and treated RS were determined according to a previously reported procedure (Teramoto et al., 2008). The quality of residue after pretreatment and RS before pretreatment was directly weighted after drying at 60°C and cooling to room temperature.

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$$\text{Residue yield} = \text{The quality of residue after pretreatment} / \text{The quality of RS before pretreatment},$$

$$\text{Cellulose yield}(\%) = \text{Cellulose content} \times \text{Residue yield},$$

$$\text{Hemicellulose yield}(\%) = \text{Hemicellulose content} \times \text{Residue yield},$$

$$\text{Lignin yield}(\%) = \text{Lignin content} \times \text{Residue yield}.$$

Their morphologies were observed via scanning electron microscopy (SEM, JSM-IT300LA; JEOL).

2.5 | Characterization of lignin and LNPs

Lignin fractions were extracted from the treated and untreated RS according to a previously reported procedure (Wen et al., 2015). The lignin was characterized via FTIR, X-ray photoelectron spectroscopy (XPS, ESCALAB 250Xi; Thermo Fisher Scientific), heteronuclear single quantum coherence (HSQC) spectroscopy, and UV–vis spectroscopy (U-4100; Hitachi). The purity of lignin was determined based on the total amount of acid-soluble lignin and acid-insoluble lignin in the samples, according to the NREL/TP-510-42618 standard procedure. The particle size and zeta potential of the LNPs at different pH values were measured using a Malvern Zetasizer Nano-ZS ZEN3600 instrument (Malvern). Helios NanoLab (Helios NanoLab G3 UC; FBI) was employed to observe the morphology of the LNPs. Moreover, DES-dissolved lignin was characterized to elucidate the formation mechanism of LNPs, and commercial alkali lignin was used as the control.

2.6 | DES recycling

The DESs were recycled via conventional organic extraction (Shen et al., 2019). For more convenient industrial applications, the recycling process was investigated in the absence of organic reagents as follows: The diluted DES treatment solution was heated in a vacuum oven at 70°C for 24 h to eliminate water after LNP separation. The samples were then weighed to estimate the recovery of DESs. To investigate the recyclability of the DESs based on these two different recycling processes, the RS bulk treatment and DES recovery experiments were repeated four times.

3 | RESULTS AND DISCUSSIONS

3.1 | Preparation and characterization of DESs

The molar ratios of the components in DESs significantly impact their freezing points. In this study, ChCl was selected as the hydrogen-bond acceptor, as it is inexpensive and has low toxicity and high biodegradability. Hydrogen bonds can be formed between the chloride ions and carboxylic acid molecules, and a stable DES can be obtained upon the complexation of one chloride ion with two carboxylic acid molecules. Thus, LA and OA were selected as the hydrogen-bond donors for the preparation of binary and ternary DESs. Opaque white crystals were only observed in two samples (3 and 4) at 60°C (Supporting Inromation: Figure S1); however, when the temperature dropped to 25°C, eight other samples solidified. Ch:1.0Lac, Ch:1.5Lac, Ch:0.5Oxa:0.5Lac, Ch:0.5Oxa:1.0Lac, and Ch:0.5Oxa:1.5Lac were completely colorless and transparent liquids. As industrial applications prefer DESs with low melting points, only the DESs that remained in a liquid state at room temperature were selected for further use.

Hydrogen bonds are the most important intermolecular interactions in DESs and are indispensable for their formation and stability (Ashworth et al., 2016). Theoretically, the length and energy of a bond and the electron density of a proton closely depend on the hydrogen bonds, which cause spectral signature variations. The hydrogen bonds in different DESs were detected via ^{1}H-NMR (Table 1 and Supporting Information: Figure S2), and the peaks at 5.65, 5.58, 5.79, 6.13, and 6.05 were attributed to the chemical shifts of hydroxyl hydrogen protons in Ch:1.0Lac, Ch:1.5Lac, Ch:0.5Oxa:0.5Lac, Ch:0.5Oxa:1.0Lac, and Ch:0.5Oxa:1.5Lac0, respectively, thus indicating the formation of hydrogen bonds between hydrogen-bond acceptors (ChCl) and hydrogen-bond donors (LA, OA, or both). Notably, Δδ(–OH) indicates the difference between the chemical shift of hydroxyl hydrogen protons in the DESs and that in ChCl. The Δδ(–OH) values of Ch:1.0Lac and Ch:1.5Lac were less than 0, thus indicating hydrogen-bond formation between ChCl and LA, which weakened the hydrogen bonding in ChCl itself. The subsequent increase in the density of the electron cloud around the hydrogen nucleus shifted the δ value of ^{1}H upfield. Moreover, the Δδ(–OH) values of Ch:0.5Oxa:0.5Lac, Ch:0.5Oxa:1.0Lac, and Ch:0.5Oxa:1.5Lac were larger than 0, indicating that the hydrogen bonds formed in the ternary DESs were stronger than those in ChCl itself. The density of the electron cloud around the hydrogen nucleus correspondingly decreased, causing the δ value of ^{1}H to shift downfield. The strongest hydrogen bonding was observed in the ternary DES Ch:0.5Oxa:1.0Lac with the highest Δδ(–OH) value (0.41). Furthermore, the FTIR spectra of the DESs (Supporting Information: Figure S3) revealed the presence of peaks at 3222, 3023, and 1719 cm^{-1}, attributed to the stretching vibrations of O–H and C–H in

TABLE 1 The chemical shifts and changes of hydroxyl hydrogen protons among different DESs.

Sample	δ (–OH)	Δδ (–OH)/ ppm	δ($OHCH_2CH_2N(CH_3)_3Cl$)	Δδ (CH_2)/ ppm
Ch:0.5Oxa:1.5Lac	6.05	0.33	3.83	0.04
Ch:0.5Oxa:1.0Lac	6.13	0.41	3.83	0.04
Ch:0.5Oxa:0.5Lac	5.79	0.07	3.83	0.04
Ch:1.5Lac	5.58	−0.14	3.83	0.04
Ch:1.0Lac	5.65	−0.07	3.83	0.04
ChCl	5.72	/	3.79	/

Abbreviation: DES, deep eutectic solvent.

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methyl and methylene and C = O, respectively (Haraźna et al., 2019). The peak corresponding to O–H stretching in the DESs exhibited a blueshift compared to that in OA. The C–O bond energy decreased, and its bond length increased, thus leading to a redshift in the peak corresponding to C = O stretching vibrations in OA after DES formation. However, compared to LA, the O–H stretching and C = O stretching vibration peaks presented a redshift and a blueshift, respectively, in the DES system, indicating that the hydrogen-bond interaction of O–H···Cl in the DES was stronger than that in OA; however, the intermolecular hydrogen bonding in LA was weakened owing to the formation of O–H···Cl in the DES. All these results confirmed hydrogen-bond formation in the DESs, and these bonds were stronger in the ternary system than in the binary system.

The viscosity, reflecting the strength of intermolecular interactions in liquids, was investigated at 60°C and 25°C (Table 2). The viscosities of DESs increased from 33.5–72.5 to 99.5–264.0 mPa/s as the temperature decreased from 60°C to 25°C. Moreover, the viscosities of the ternary DESs were significantly lower than those of the binary DESs. Generally, hydrogen bonding is positively related to the viscosity (Wang et al., 2019); however, this was not observed in the studied DESs. Furthermore, the viscosities of the obtained DESs were significantly lower than that of edible oil (50–100 mPa/s). These excellent properties indicate the applicability of DESs in industrial applications.

The α and β values of the binary DESs were 1.16–1.22 and 0.37, respectively (Table 3). In the ternary DESs, the α values significantly increased to 1.32–1.34, whereas the β values decreased to 0.15–0.27. Notably, DESs with high α and low β values indicate strong hydrogen bonding, owing to the high hydrogen-bond acidity and low hydrogen bond basicity. In fact, high hydrogen-bond acidity can promote the destruction of lignocellulose and the dissolution of lignin (Liu et al., 2017). Consequently, ternary DESs with high α and low β values demonstrate the greatest potential for lignin dissolution and separation. Furthermore, the π values of the obtained DESs were not significantly different, indicating that the polarities of all the obtained DESs may be relatively similar.

TABLE 2 The viscosity of different DESs at 60°C and 25°C.

Sample	Viscosity at 60°C/ mPa/s	Viscosity at 25°C/ mPa/s
Ch:0.5Oxa:1.5Lac	33.5	99.5
Ch:0.5Oxa:1.0Lac	35.5	117.0
Ch:0.5Oxa:0.5Lac	49.0	161.0
Ch:1.5Lac	48.0	169.5
Ch:1.0Lac	72.5	264.0
Lactic acid	17.0	50.5

Abbreviation: DES, deep eutectic solvent.

TABLE 3 The ENR and Kamlet–Taft parameters of different DESs.

Sample	E_{NR}	α	β	π
Ch:0.5Oxa:1.5Lac	44.81	1.32	0.27	1.09
Ch:0.5Oxa:1.0Lac	48.13	1.34	0.15	1.06
Ch:0.5Oxa:0.5Lac	47.81	1.32	0.22	1.09
Ch:1.5Lac	47.81	1.22	0.37	1.13
Ch:1.0Lac	47.81	1.16	0.37	1.13
Lactic acid	47.97	1.31	0.25	1.06

Abbreviation: DES, deep eutectic solvent.

3.2 | RS fractionation

To verify the fractionation efficiency of the DESs, a microwave-assisted treatment of the bulk RS was conducted for a duration of 3 and 4 min. The obtained solid–liquid mixtures were poured into petri dishes, and the colors of all the solid–liquid mixtures turned relatively light brown after 3 min of treatment with the binary DESs (Figure 1a), thus indicating that some RS was fractionated, and a small amount of lignin was dissolved in the DESs; in contrast, after treatment with ternary DESs, a large amount of RS was broken down, and the mixture color turned dark brown. Upon increasing the treatment time to 4 min (Figure 1b), the rate of dissolution by the binary DESs improved; however, some RS remained undissolved, whereas after treatment with the ternary DESs, most of the RS was dissolved, and the color of the mixture turned darker. The mixture was further subjected to solid–liquid separation, and the resulting solid residue was washed to a neutral pH and then freeze-dried. Bulk RS was observed in all samples treated with binary and ternary DESs for 3 min (Figure 1c). After 4 min of ternary DES treatment, the RS changed from bulk to powder and agglomerated into pieces, while some bulk RS was still present in the residue of the binary treatment (Figure 1d). Thus, the dissolution rates of lignin treated by the ternary DESs were better than those treated by the binary DESs. Furthermore, the SEM images (Supporting Information: Figure S4) of different RS samples revealed that the surface of the untreated RS was dense, rough, and completely covered with small particles, whereas the surface of the RS treated by the ternary DESs was loose, smooth, and porous, primarily owing to the complete removal of particles; in contrast, only minor changes were observed on the surface of the RS after treatment with the binary DESs. This further confirmed that ternary DESs demonstrated a better ability to decompose lignocellulose and enabled more lignin dissolution from the RS.

The lignin separation efficiencies of the DESs were evaluated by observing the changes in the RS components before and after treatment (Figure 2). The contents of cellulose, hemicellulose, and lignin in the untreated RS were 45.2%, 39.8%, and 10.8%, respectively, while ash and other substances accounted for 4%–5%. The RS components were well separated under microwave conditions, and in particular, the lignin removal rate upon treatment with ternary DESs for 4–5 min was as high as 60.8%–70.3%; however, the treatment with binary DESs under the same conditions dissolved and separated only 7.6%–36.7% of lignin (Figure 2a,b). This further proved that ternary DESs exhibited better

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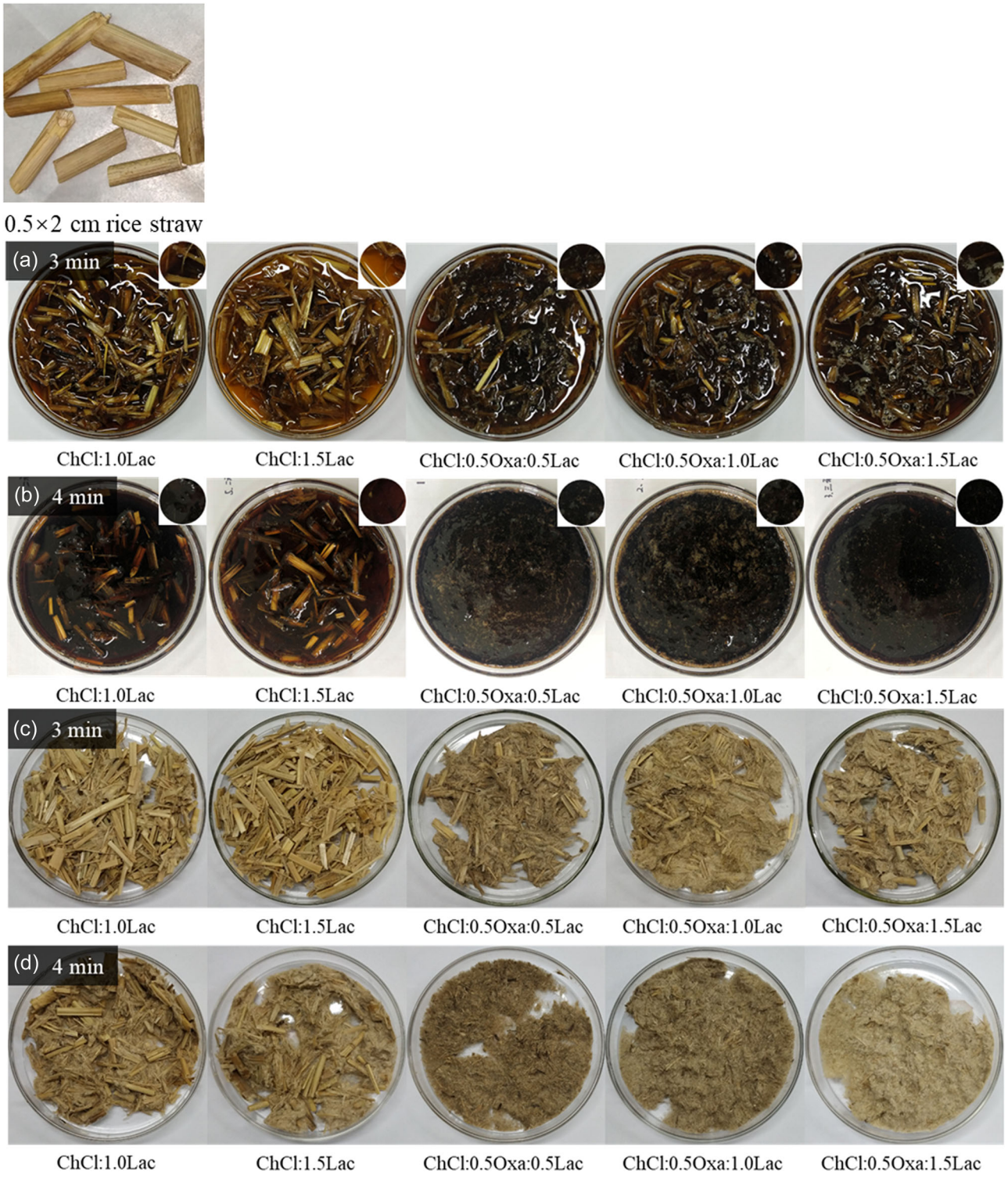


FIGURE 1 Rice Straw pretreated by various DESs under different conditions. (a) The mixtures obtained after pretreatment for 3 min; (b) The mixtures obtained after pretreatment for 4 min; (c) The residues obtained after pretreatment for 3 min; (d) The residues obtained after pretreatment for 4 min.

lignocellulose dissolution abilities compared to binary DESs. As the reaction time increased from 4 to 5 min, the lignin separation effect imparted by the ternary DES treatment did not significantly improve. Therefore, the optimal duration for lignin separation using ternary DESs was 4 min. Furthermore, the ternary DES treatment for 4–5 min significantly removed hemicellulose from the RS, and the removal rates (71.4%–80.3%) were much higher than those (32.3%–57.3%) for the binary DESs (Figure 2c). Lignin and hemicellulose removal using ternary DESs not only weakened the resistant structure of the RS but also increased the cellulose content from 45.2% to 59.4%–64.7%

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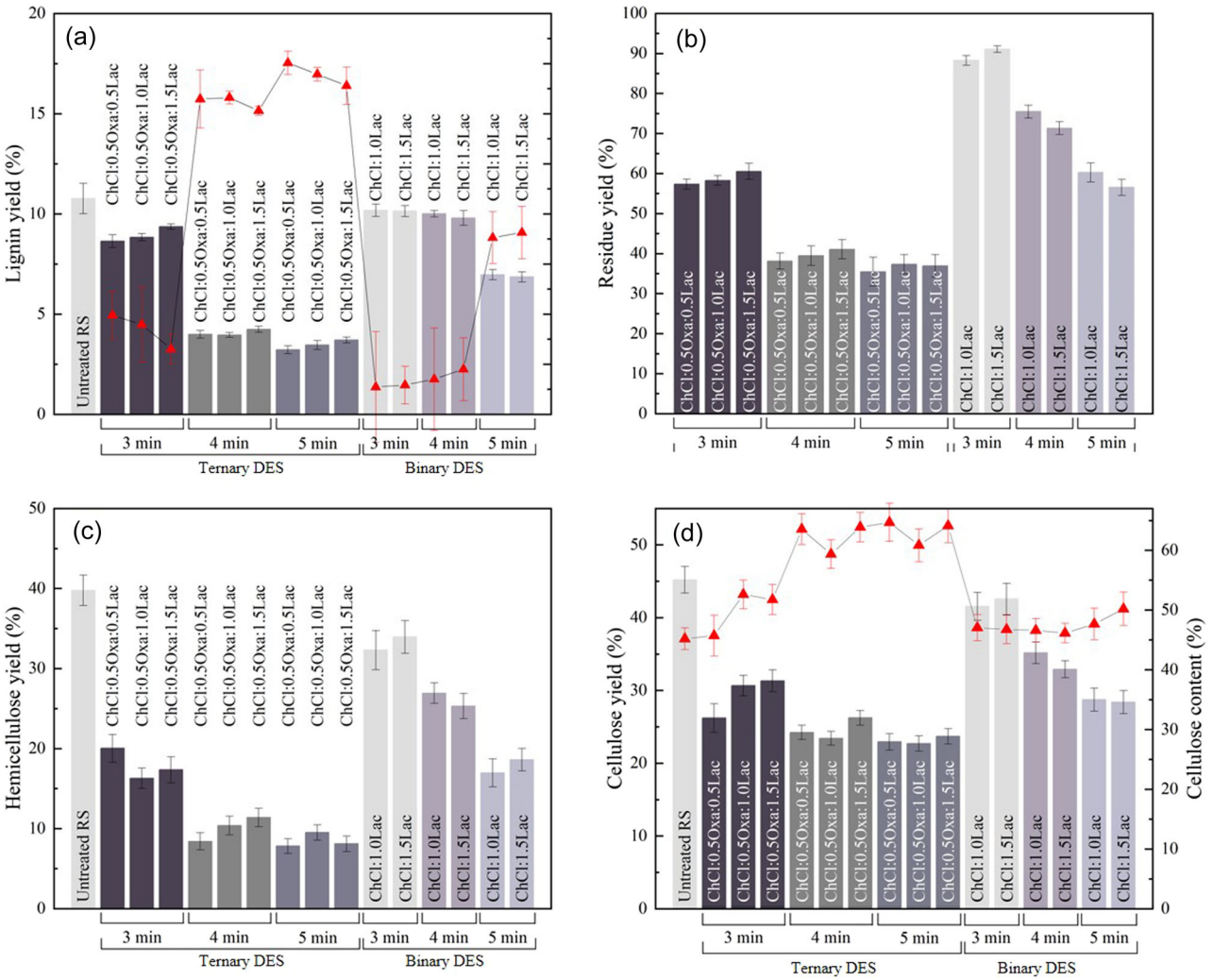


FIGURE 2 The fractionation of rice straw by different DESs under various conditions. (a), Lignin yield; (b), Residue yield; (c), Hemicellulose yield; (d), Cellulose yield. DES, deep eutectic solvent.

(Figure 2d). Upon extending the reaction time, the residue yields of ternary and binary DES treatments decreased as more cellulose, hemicellulose, and lignin were transformed into small soluble molecular compounds, which enhanced the reaction rates. The above results indicate that microwave-assisted treatment with ternary DESs for 4 min resulted in better dissolution of the RS components. The lignin removal rate was in the range of 60.8%–63.4%, and the highest rate was obtained with the ternary DES Ch:0.5Oxa:1.0Lac, resulting in relatively pure cellulose. Notably, fiber materials with a higher cellulose content generally contribute to a high reducing sugar yield, indicating that this strategy may realize the effective utilization of all components in lignocellulose. However, further related studies are currently underway. Under the same conditions, a lignin removal rate of only 7.6%–9.6% was obtained by the treatment of RS using binary DESs.

In previous studies, lignocellulose powder (20–80 mesh, 180–800 μm) was employed as the starting material for pretreatment or treatment using DESs; the corresponding processes involved a grinding procedure using a mill and a subsequent sifting procedure using a sifter with 20–80 meshes (Alvarez-Vasco et al., 2016; Ji et al., 2020; Liu et al., 2018; Luo et al., 2020). A delignification proportion higher than 50% was obtained; however, multiple mechanical procedures are generally time and energy consuming. The results of this study indicate comparatively superior delignification. Significantly, the RS was cut into small bulks (0.5 × 2.0 cm) and directly employed as the feedstock in the microwave-assisted ternary DES system (Figure 1). If the volume of the treatment system is sufficiently large, the RS can be treated directly without any mechanical processes. Such an energy-saving strategy is highly appealing in industrial applications.

3.3 | Lignin depolymerization and aggregation

The structural evolution of lignin from the Ch:0.5Oxa:1.0Lac treatment (the highest lignin removal rate was obtained) was analyzed using 2D-HSQC NMR and FTIR. Notably, the major

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restriction to lignin fractionation is the strong recalcitrance of lignin–carbohydrate complexes (LCCs), which consist of a network of covalent bonding and hydrogen-bond interactions (Tarasov et al., 2018). Compared with the untreated RS, the cross-signals of X2, X3, and X4 disappeared from lignin in the RS after DES treatment (Figure 3a,b, and Supporting Information: Table S2), thus indicating the cleavage of certain linkages between xylan and lignin in the LCCs. Moreover, some signals corresponding to specific bonds for the leakage of basic structures in lignin, including β–O–4 (A_β(G/H) and A_β(S)), β–β (B_γ), and β–5 (C_β and C_γ), almost disappeared after the treatment (Figure 3c,d, and Supporting Information: Table S2), owing to the destruction of ether bonds in these structures. Other signals related to the *p*-hydroxycinnamyl alcohol end-group (I_γ) and *p*-coumarate ($PCE_{2,6}$) completely disappeared, while a new signal of

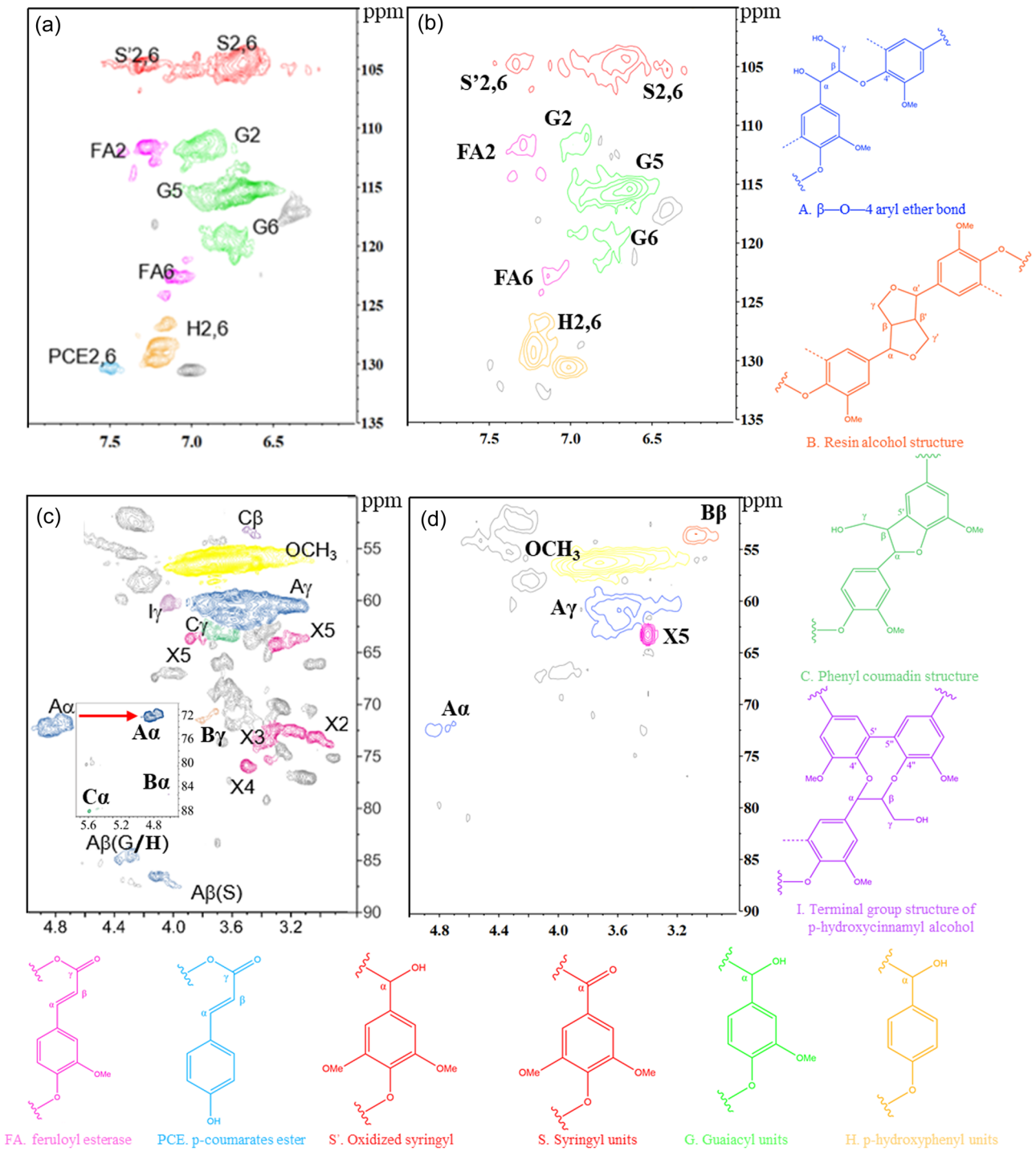


FIGURE 3 Heteronuclear single quantum coherence spectrum of untreated rice straw (a,c) and DES (ChCl:0.5Oxa:1.0Lac-4min) treated rice straw (b,d). (a,b) aromatic ring region; (c,b) side chain region).

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B_β corresponding to the resinol structure appeared in the HSQC spectra, indicating the occurrence of lignin condensation during the treatment. Notably, the syringyl/guaiacyl (S/G) ratio is an important index for verifying changes in the lignin structure. The initial lignin in the RS bulks exhibited a typical S-G-type structure with an S/G ratio of 1.28 (Supporting Information: Table S3). Multiple linkages including β–O–4 (86.3%), β–β (6.4%), and β–5 (7.3%) connected the monolignol units to form the lignin. The S/G ratio of lignin in the RS significantly increased from 1.28 to 2.76 after DES treatment, indicating that G-type lignin primarily underwent fractionation during the treatment. Consequently, the dissociated lignin possessed increased numbers of G units. Furthermore, the C–C bonds in β–β and β–5 disappeared in the dissociated lignin, and only cross-signals associated with β–O–4 were detected in the lignin of RS treated with the DES. The FTIR spectra (Supporting Information: Figure S5) revealed the presence of peak at 1238 cm^{-1}, attributed to the carboxyl groups and the C–O stretching in ester groups. This peak disappeared after the DES treatment, confirming the cleavage of benzyl esters in xylan. Notably, in a DES system, hydrogen-bond interactions may be weakened owing to competing hydrogen-bond formation processes between the chloride ions of the DESs and hydroxyl groups in the LCCs, thus motivating lignin fractionation. Therefore, the DES treatment altered the structures of LCCs and cleaved β–O–4 to fractionate G-type lignin in the RS.

A homogeneous opaque liquid was obtained upon the addition of deionized water to the filtrate obtained from the treatment of RS with the ternary DES CHCl:0.5oxa:1.0lac for 4 min. Numerous suspended particles were observed in the solution, and the Dindal phenomenon was observed upon infrared irradiation (Figure 4a), indicating that a colloidal solution was obtained after dilution of the filtrate, and the particle size of the suspended particles in the solution was extremely small. The SEM analysis confirmed that the obtained LNPs were well dispersed (Supporting Information: Figure S6). The particle size distribution of the LNPs was relatively concentrated, and the particle sizes ranged from 48 to 95 nm (Figure 4b). Furthermore, the high purity of the LNPs is extremely important for their applications. The highest purity of LNPs in our study was 86.8 ± 0.5% without ethanol purification, which was comparable to those reported in previous studies (Ai et al., 2020; Luo et al., 2020; Lynam et al., 2017; Wang et al., 2020).

3.4 | High stability of the regenerated LNPs

In general, C–C bonds are more resistant than β–O–4 ethers. Notably, G-type lignin contains more resistant bonds, such as the β–5 bonds and β–β bonds, compared with S-type lignin, owing to the availability of the C5 position for coupling. The significant signals of $S_{2,6}$, $S'_{2,6}$, G_2, G_5, G_6, and $H_{2,6}$ observed in the aromatic ring region of the HSQC spectra confirmed that the obtained LNPs contained typical S, G, and *p*-hydroxyphenyl lignin structures (Figure 5a and Supporting Information: Table S2). These basic units were primarily connected via numerous C–C linkages, including β–5 and β–β bonds, as confirmed by the appearance of C_γ (β–5) and B_β (β–β); the appearance of A_γ indicated that some β–O–4 bonds were still present in the LNPs (Figure 5b and Supporting Information: Table S2), strongly indicating that the LNPs exhibited high structural stability for composite material preparation and applications. In practical applications, a significant challenge with regard to nanoparticle dispersions lies in retaining stability at various pH values. Generally, LNPs are negatively charged owing to the carboxyl and/or hydroxyl groups. In our analysis, functional groups of the LNPs were also detected (Supporting Information: Figure S7). The absorption bands at 3361 and 1363 cm^{-1} in the FTIR spectrum were attributed to the hydroxyl stretching vibration and phenolic OH, respectively, thus indicating the polyhydroxy structure of the LNPs. Furthermore, the C = O stretching vibration (unconjugated at 1647 cm^{-1}) verified the absence of carboxyl groups in the LNPs. These hydroxyl and carboxyl groups may promote the formation of electrical double layers, consequently stabilizing nanoparticle dispersion via electrical double-layer repulsion. The stability of LNPs was investigated by measuring the zeta potential values in a pH

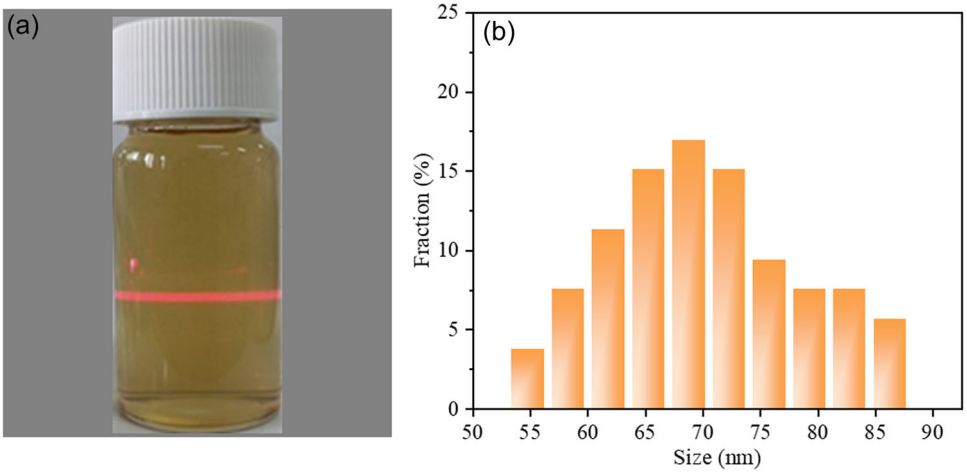


FIGURE 4 Properties of lignin nanoparticles in solution. (a), Tyndall phenomenon of colloidal solution; (b), particle size distribution of lignin nanoparticles.

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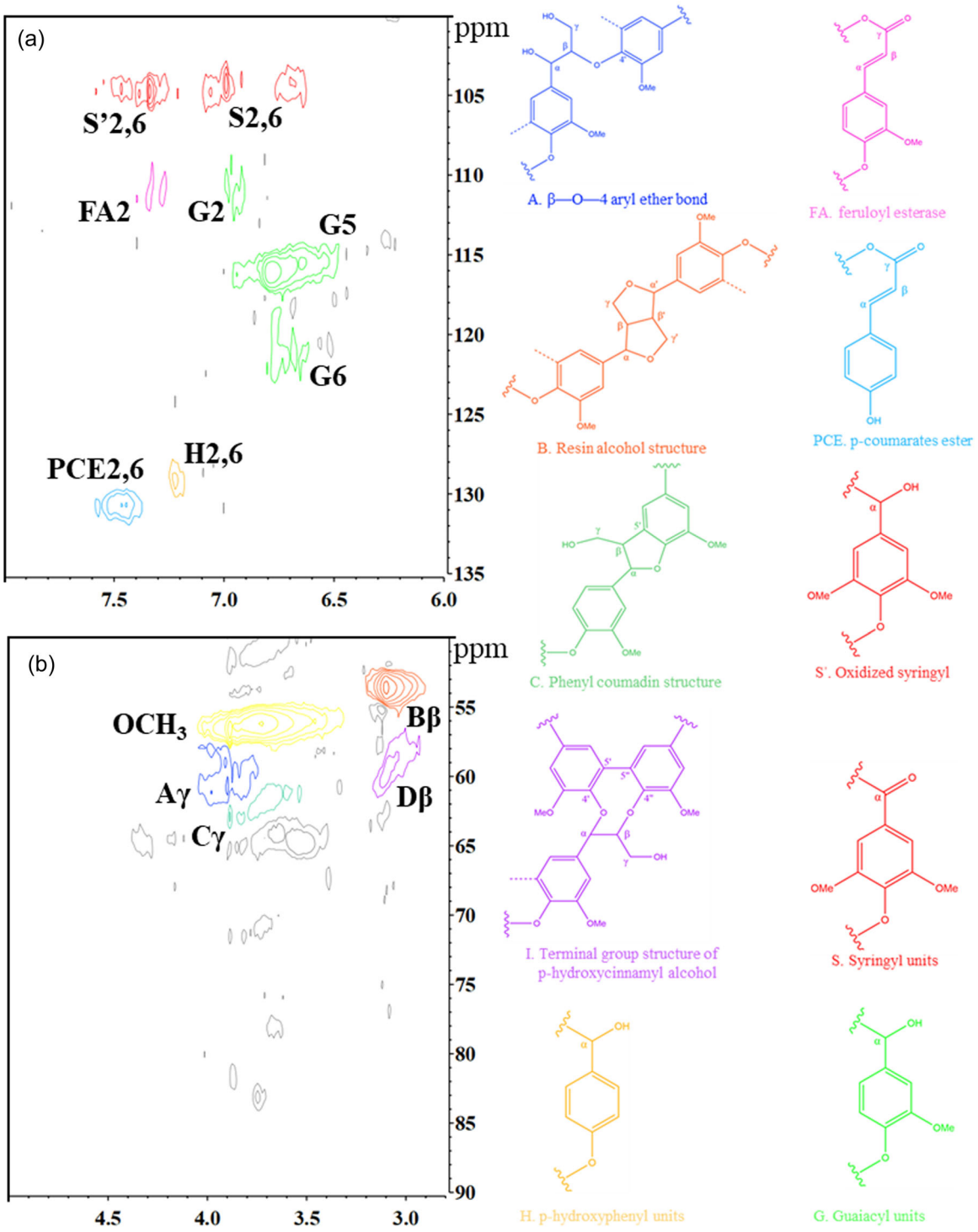


FIGURE 5 Heteronuclear single quantum coherence spectrum of lignin nanoparticles. (a) aromatic region; (b) side chain region.

range of 4–12 (Figure 6a). The zeta potential values of the LNPs ranged from − −25.5 to − −17.1 MV, thus indicating good colloidal stability of the LNPs. Furthermore, the thermal stability of the LNPs was analyzed. At a temperature lower than 220°C, only a small portion of the LNPs was pyrolyzed (Figure 6b). The maximum decomposition temperature of the LNPs was higher (Td: 358.0°C) than that of alkali lignin (Td: 322.5°C), thus indicating good thermal stability of the obtained LNPs (Figure 6c).

3.5 | Formation mechanism of LNPs

Lignin is an amphiphilic polymer consisting of both hydrophilic (e.g., alcoholic hydroxyl groups) and hydrophobic groups (e.g., aromatic rings), which enables the dissolution or aggregation of lignin in different media (Heitner et al., 2016). Lignin aggregation can be classified into two ways. The first is the molecular aggregation of

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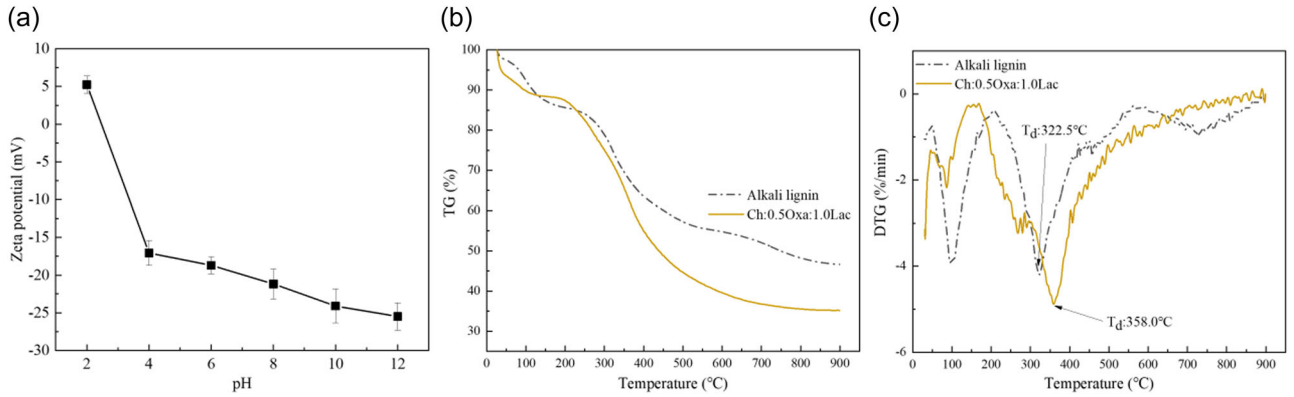


FIGURE 6 Properties of lignin nanoparticles. (a) Effect of pH on Zeta potential of lignin nanoparticles; (b) TG curve of lignin nanoparticles; (c) DTG curve of lignin nanoparticles.

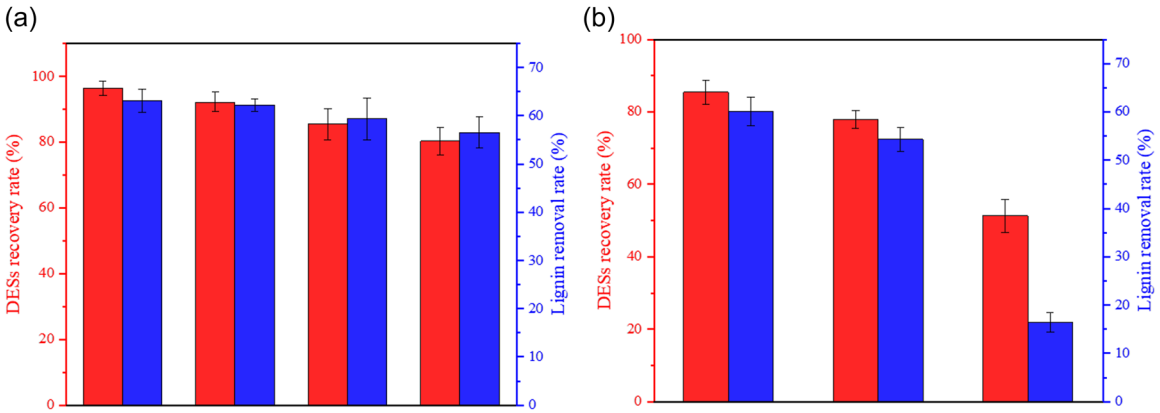


FIGURE 7 The recovery of DES_s with (a) and without (b) organic process. DES, deep eutectic solvent.

polymer chains due to van der Waals attraction and/or hydrogen bonds and the second is the π–π interaction of the aromatic groups owing to nonbonded orbital interactions (Dhotel et al., 2013). The XPS analysis revealed that the lignin considered in this study demonstrated a high carbon-to-oxygen ratio (C/O = 3) and lacked carbon water chains (Supporting Information: Figure S8 and Table S4). Consequently, it was speculated that the primary factors affecting lignin aggregation were not van der Waals forces but π–π interactions. To prove the occurrence of hydrolysis, FTIR and UV–vis analyses were performed. As depicted in Supporting Information: Figure S9a, a red shift was observed in the typical lignin benzene ring absorption peak from 1513 cm^{-1} for alkali lignin to 1480 cm^{-1} for the dissolved lignin. This phenomenon was attributed to the strong π–π stacking interactions (Qian et al., 2014). When LNPs were formed through the colloidization of the dissolved lignin, the absorption peak at 1480 cm^{-1} shifted back to 1513 cm^{-1} owing to the weakening of the π–π stacking interaction. Similarly, in the UV–vis spectrum (Supporting Information: Figure S9b), the typical lignin benzene ring absorption peak of alkali lignin at 280 nm blue-shifted to 271 nm for the dissolved lignin, which resulted from π–π stacking interactions. By contrast, the absorption peak at 271 nm red-shifted back to 277 nm owing to the weakening of π–π stacking interactions after the formation of LNPs from dissolved lignin (Lee et al., 2013). Therefore, the formation of LNPs with the lignin obtained from RS treated with DESs is driven by π–π stacking interactions.

3.6 | DES recycling

The recycling of DESs is important for reducing the cost of LNP preparation. Thus, DES recycling was performed using two different methods. The recycling performance was evaluated based on the DES recovery and lignin removal rates. As depicted in Figure 7a, 80.2%–96.3% of DESs were recovered in each recycle. The next round of RS pretreatment with the microwave system was performed using the DESs recovered via an organic extraction process. The lignin removal rate decreased with an increase in the DES recycling number, which was primarily owing to the impurities present in the DESs after each recycling stage (Qiu & Aita, 2013). The recovered DESs after the fourth recycling stage were still able to remove 54.5% of lignin from the RS. Without organic extraction, more than 70% of the DESs were recovered after each of the first two recycling stages

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(Figure 7b). However, the recovery rate rapidly decreased to 51.3% after the third recycling stage. Furthermore, 60.6% and 53.3% lignin removal rates were obtained after the first and second recycling stages, respectively; however, a significant decline (16.4%) was observed after the third cycle. A certain amount of lignin remained in the recovered DESs and could not be removed without organic extraction, which resulted in a low efficiency of DESs for lignocellulose pretreatment. Thus, DESs can be efficiently recovered using the process involved in organic extraction and reused in the RS pretreatment. Notably, the rates of DES recovery and lignin removal after the first two rounds of DES recycling without organic extraction were comparable. Therefore, it is speculated that significant cost reduction may be realized if the two methods are combined. However, further studies are required to develop new DES recycling strategies.

4 | CONCLUSION

In conclusion, this paper reports the preparation of LNPs via ultrafast (4 min) microwave-assisted fractionation of RS without grinding using ternary DESs. The proposed method allowed 63.4% of lignin removed from RS to prepare LNPs with a high lignin purity of 86.8% and small average particle size ranged from 48 to 95 nm. The LNPs aggregated with dissolved lignin via π–π stacking interactions. Thus, this study provides a low-cost and green strategy for lignin valorization, which may promote lignocellulose pretreatment and the separation of its components.

AUTHOR CONTRIBUTIONS

Zhiyan Yan: Data generation; manuscript preparation; manuscript editing. **Zhongren Wang**: Resources; data analysis; investigation. **Yuehui Chen**: Resources; visualization; manuscript editing. **Chenrui Liu**: Data generation; experimental design; data analysis. **Yun Liu**: Conceptualization; manuscript editing; funding acquisition. **Ruijie Li**: Manuscript preparation; resources. **Mengying Si**: Conceptualization; experimental design; data generation. **Yan Shi**: Manuscript preparation.

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DATA AVAILABILITY STATEMENT

Research data are not shared.

ORCID

Yun Liu http://orcid.org/0000-0002-1553-0798

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